script.md — per-member speaking blocks (C6)

Talk length: TBD-C6 (not supplied); timed for a working assumption of **10:00**, with the cut list bringing 10:20 spoken under it. **Team size: FIVE (confirmed): Chao Hsien, Jiwan Kang, Ng Siu Hin, Su Wei-Chang, Tina Tien.** Blocks M1–M5 are contiguous and sized 1:45–2:55; every member speaks (rubric requirement). Assigned in roster order below; M4 (Su Wei-Chang) should be the strongest English speaker — it carries the AQC win, the trap, and the hardware failure; swap M4 with another block if that is not Su Wei-Chang. English comfort per member: TBD-C6. Divider pages: advance silently.

Numbers rule: every number below is grep-verified against RESULTS.md and tagged. Do not improvise numbers on stage.

Pronunciation: *ancilla* an-SIL-la · *Hadamard* HA-da-mar · χ "kai" · *Loschmidt* LOSH-mit · *ansatz* AHN-zahts · *AQC* ay-cue-see · *Trotter* TROT-ter · *Fermi–Hubbard* FER-mee HUB-bard.

Block M1 — open (S1–S3) — Chao Hsien — 1:50

> We are Team 8, the Garbage Collectors. This is the shadow-enhanced Hadamard test — and our story is how we made it scale, found the trap that silently breaks it, and then tested our own claim on real hardware. Including the part where the hardware said no.
>
> Four numbers up front. Thirty-six times fewer gates — as a compilation result [R046]. One single-qubit gate to fix an error larger than the signal [R046]. Two quantum processors where our own pre-registered prediction failed by twelve times [R054]. And one identity that is exactly zero, which hands us a device-error bar with no extra circuits [R044].
>
> The challenge had five tracks. We completed all five — the mandatory one at twelve out of twelve seeds [R019]. This talk is Track B, the one we took furthest.

Block M2 — the question and the instrument (S4–S6) — Jiwan Kang — 2:05

> Here is the question everything hangs on. Every real simulation runs a cheap approximation W instead of the true evolution U. How wrong is W — and wrong in *what*? Classically you simulate both. That is exactly what you cannot do at scale. Track B answers it on the device.
>
> One change to the textbook circuit. Two X gates around the controlled-W make W fire when the ancilla is zero, and U when it is one. The ancilla now interferes two dynamics against each other. Nothing downstream changes.
>
> A new circuit gets its own validation. The identity holds to ten to the minus fourteen on two models [R034, R042] — and that check caught a real endianness bug before it ever reached a quantum processor [R042].

Block M3 — shadows and the honest rivals (S7–S9) — Ng Siu Hin — 1:45

> For anyone new to shadows: measure each qubit in a random basis, keep the record. One record set then estimates every Pauli observable at once, at a variance of three to the weight — a model we verified to within two percent [R029]. And here is our completion of Track B: delete the final Hadamard, and the shadows split into the sum and the difference of the two dynamics. Add and subtract — you get every observable under W and under U separately [R042]. The ancilla says how much W is wrong. The garbage says which observable.

Due diligence: the literature already verifies compiled circuits — the Loschmidt echo, the Hilbert–Schmidt test. We implemented both, verified both [R043]. And the scoreboard is not flattering: the echo is four to six times cheaper at every size we measured [R043]. If you want one number, use the echo. Our arm exists for what nothing else returns: the phase, and the per-observable breakdown.

Block M4 — the win, the trap, the verdict (S10–S12) — Su Wei-Chang — 2:55

Now the bottleneck. The standard construction of the controlled evolution costs four to the n-plus-one gates. AQC grows roughly linearly. They cross at four qubits; by seven, AQC is thirty-six times cheaper at infidelity three times ten to the minus four [R046]. That survives a 2D lattice [R049], the Fermi–Hubbard model [R051], and real device routing — which widens the gap [R052].

But shipping it naively destroys the experiment. The compiler maximises a magnitude; a Hadamard test measures a phase. The compression converges beautifully and returns chi wrong by up to three radians — an error as large as the signal [R046]. The fix is one phase gate on the ancilla: error one point three three down to zero point zero zero eight [R046]. We met the same magnitude-hides-phase failure two more times — once in a teammate's hardware data [R053], and once, as you will now see, in our own.

We tested our own claim. Three constructions, one job, two devices, and the survival numbers written down before submitting. Every arm came back as noise — ours twelve times below its own prediction [R054]. And the deepest, deadest circuit got the best score: survival one point four eight five. That is unphysical — chi cannot exceed one. It is a relaxation floor, and a magnitude-only metric crowns it the winner [R054]. We know exactly what error rate changes this verdict; the arithmetic is in appendix A-seven [R055].

Block M5 — hardware close and claims (S13–S17) — Tina Tien — 1:45

Track B itself, same discipline: the two-gate echo beat our arm twenty-seven times on hardware — confirmed on an independent run when we extended the time window [R044, R059]. One clean win survives: a charge difference that must be exactly zero, so its measured value is pure device error — an error bar with no reference state and no simulation [R044].

What we claim: the compilation result, the trap and its one-gate fix, the completion of Track B, and the hardware tests themselves. What we do not claim: that the gate-count win reaches today's hardware — we showed it does not [R054] — and nothing against classical computation; the benchmark is small on purpose, so every number is graded against truth.

One future direction, already verified: the same circuit with a maximally mixed input becomes a density-of-states estimator [R045].

Everything is public behind this QR — a sixty-two-row results ledger, the bug log including our own retractions, and the scripts behind every hardware number. [Team slide: each of the FIVE members states their part — Chao Hsien, Jiwan Kang, Ng Siu Hin, Su Wei-Chang, Tina Tien.]

[Summary slide, ~10 s:] To summarise: five tracks completed. A thirty-six-times compilation win. A trap found — and fixed with one gate. And an honest hardware no, predicted in advance. Every line here has its source row. We leave this up — thank you, questions welcome.

Timing (assumes 10:00 slot — TBD-C6)

block	speaker	time	cumulative
M1 open (S1–S3)	Chao Hsien	1:50	1:50
M2 question+instrument (S4–S6)	Jiwan Kang	2:05	3:55
M3 shadows+rivals (S7–S9)	Ng Siu Hin	1:45	5:40
M4 win/trap/verdict (S10–S12)	Su Wei-Chang	2:55	8:35
M5 hardware close (S13–S17 + summary)	Tina Tien	1:55	**10:30**

10:30 spoken against 10:00 is deliberate: the cut list brings it under with the first two items (or cut 1 plus a brisk M1). **Cuts, in order:** (1) S15 DOS sentence −0:20; (2) shadow primer half of S7 −0:40, only for an expert room; (3) merge S8 into S9 −0:30. **Never cut:** S11 the trap, S12 the failure, S14 the claims. Checkpoint: start Block M4 by 5:45 or apply cut 1 immediately. All five members also appear on the team slide.